小白程式入門
電子書

從零開始
用故事帶你踏入程式世界

- 10 個章節完整學習路徑
- 真實故事引導入門
- 從觀念到實作一次搞懂

```
def learn():
    start()
    practice()
    grow()
```

桑尼資料科學
Data Sunnie

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■■■■■■■■■■■■■■■■■

■■■■■■■■■ | Data Sunnie

■■

PART 1■■■■■

■■	■■	■■■■
01	■■■■■■■■	■■■■■■■■■■■■
02	■■■■■■■	■■■■■■■■■■■■■
03	■■■■■■■	■■■■■■■■■■■

PART 2■■■■■

■■	■■	■■■■
04	■■■■■	■■■■■■■■■■■
05	■■■■■■■	■■■■■■■■■
06	■■■■■■■■■	■■■■■■■■■■■

PART 3■■■■■

■■	■■	■■■■
07	■■■■■■■■■	■■■■■■■■■
08	■■■■■■■	■■■■■■■■■
09	■■■■■■■■■	■■■■■■■■■■■
10	■■■■■■■■■	■■■■■■■■■

■■■■■

■■■■■■■■■■■■■■■■■■■■■■■■■■■■■

■■■■■■■■■■■■■■■■■■■■■■

■■■■■■■■■■■■■■

?

CHAPTER 01

程式到底是什麼？

打破迷思，理解程式的本質

01

1

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]35[redacted]——[redacted]

[redacted]

[redacted]35[redacted]...

[redacted]——[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]——[redacted]

[redacted]

[redacted]——[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted]

[redacted]——[redacted]

[redacted]

[redacted]

```
[redacted]
```

[redacted]Google Maps[redacted]→[redacted]→[redacted]

[redacted]

[redacted]

[redacted]

[redacted]——[redacted]90%[redacted]3D[redacted]

[redacted]

[redacted]50[redacted]if[redacted]for[redacted]while[redacted]true[redacted]false[redacted]

[redacted]

[redacted]35[redacted]

[redacted]81[redacted]

[redacted]30[redacted]40[redacted]50[redacted]

[redacted]

[redacted]

[redacted]Hello World[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]——[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]——[redacted]

[redacted]

[redacted]

[redacted]

[redacted]——[redacted]

[redacted]

[redacted]——[redacted]

[redacted]

[redacted]

[redacted]

[redacted] [redacted]

CHAPTER 02

程式語言大觀園

認識各種語言，找到適合你的

02

■2■■■■■■■■■

■■■■■■■■■■■■■■

■■■■■■■■■■■■■■

[redacted]

[redacted]

[redacted]——[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted]Hello, [redacted]

Python[redacted]

```
name = "[redacted]"
print(f"Hello, {name}")
```

JavaScript[redacted]

```
let name = "[redacted]";
console.log(`Hello, ${name}`);
```

Java[redacted]

```
String name = "[redacted]";
System.out.prinfln("Hello, " + name);
```

[redacted]

Python[redacted]

[redacted]Python[redacted]——[redacted]

[redacted]Python[redacted]

```
[redacted]
[redacted] = 75
[redacted] = 65

if [redacted] > [redacted]:
    print("[redacted]")
else:
    print("[redacted]")
```

Python[redacted]

Python[redacted]

- [redacted]
- [redacted]1000[redacted]
- [redacted]
- [redacted]

[redacted]Python[redacted]Excel[redacted]——[redacted]

JavaScript[redacted]

[redacted]Python[redacted]JavaScript[redacted]——[redacted]

[redacted]JavaScript[redacted]

[redacted]JavaScript[redacted]

```
// [redacted]
[redacted].addEventListener('click', function() {
    if ([redacted] === "[redacted]") {
        [redacted].style.color = "[redacted]";
        [redacted]('[redacted]');
    } else {
        [redacted].style.color = "[redacted]";
        [redacted]("[redacted]");
```

JavaScript[redacted]

JavaScript[redacted]

- [redacted]
- [redacted]
- [redacted]——[redacted]JavaScript[redacted]
- [redacted]App[redacted]React Native[redacted]JavaScript[redacted]App[redacted]

[redacted]JavaScript[redacted]

[redacted]

Java[redacted]

[redacted]Java[redacted]

[redacted]Java[redacted]

Java[redacted]

Go[redacted]

[redacted]Go[redacted]

[redacted]Go[redacted]

Go[redacted]

Swift[redacted]

[redacted]iOS App[redacted]Swift[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted] → JavaScript [redacted]

[redacted] → Python [redacted]

[redacted] → Java [redacted]Java[redacted]

[redacted]**iOS App** → Swift [redacted]

[redacted] → C [redacted]

[redacted]

[redacted]3-5[redacted]

[redacted]Python[redacted]JavaScript[redacted]

[redacted]

[redacted]

[redacted]Python[redacted]

[redacted]Python[redacted]C[redacted]Instagram[redacted]Python[redacted]Python[redacted]

[redacted]JavaScript[redacted]

Netflix[redacted]PayPal[redacted]Uber[redacted]JavaScript[redacted]

[redacted]Java[redacted]

[redacted]900[redacted]Java[redacted]Android App[redacted]Java[redacted]

[redacted]C++[redacted]

[redacted]C++[redacted]3A[redacted]C++[redacted]

[redacted]

[redacted]

[redacted] [redacted]

[redacted]

[redacted]Python[redacted]JavaScript[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]Python[redacted] [redacted]JavaScript[redacted]
[redacted]App[redacted]iPhone[redacted]Android[redacted]

[redacted]

[redacted]

[redacted]Hello[redacted]

[redacted]

[redacted]

[redacted]its own beauty[redacted]

[redacted]

[redacted]——[redacted]

[redacted]

[redacted]

[redacted]

[redacted]*The best programming language is the one you actually use.*[redacted] [redacted]

CHAPTER 03

像電腦一樣思考

培養運算思維與邏輯能力

03

3

[redacted]

[redacted]*Python*[redacted]

[redacted]

[redacted]VIP[redacted]

[redacted]Python[redacted]

[redacted]

[redacted]

[redacted] [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

```
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted] > 10000[redacted]VIP
[redacted]
```

[redacted]

[redacted]

1. [redacted]
2. [redacted]
3. [redacted]
4. [redacted]
5. [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

```
[redacted]()
[redacted]()
[redacted]()
[redacted]VIP([redacted])
```

[redacted]

[redacted]——[redacted]

[redacted]...[redacted]...[redacted]...[redacted]

[redacted]

```
┌─ 1. [redacted]
│  [redacted] = [redacted]([redacted])
│
├─ 2. [redacted]
│  [redacted] = [redacted]([redacted])
│
├─ 3. [redacted]
│  [redacted] = [redacted]([redacted])
│
└─ 4. [redacted]
   [redacted]VIP([redacted])
```

[redacted]

[redacted]...[redacted]...[redacted]...

[redacted]

[redacted]

[redacted]VIP[redacted]
[redacted]10000[redacted]VIP[redacted]

[redacted]

- [redacted]→[redacted]10000[redacted]
- [redacted]VIP[redacted]→[redacted]

[redacted]

[redacted]VIP[redacted]9[redacted]...[redacted]

[redacted]

```
[redacted]
[redacted]1000[redacted]
[redacted] → [redacted]166[redacted]
[redacted]VIP[redacted]
[redacted] → [redacted]9[redacted]
[redacted] → [redacted]
[redacted] → [redacted]
[redacted]
[redacted] → [redacted]
[redacted] → [redacted]
```

[redacted]

[redacted]

[redacted]100[redacted]1[redacted]2[redacted]3[redacted]4[redacted]5...[redacted]100[redacted]1[redacted]100[redacted]

[redacted]

[redacted]

1. [redacted]
2. [redacted]
3. [redacted]

[redacted]1000[redacted]

```
[redacted]1[redacted]
[redacted]2[redacted]
[redacted]3[redacted]
[redacted]1666[redacted]
```

[redacted]

```
for [redacted] in [redacted]:
    [redacted]([redacted])
```

[redacted]

[redacted]

```
while [redacted]:
    [redacted]
    # [redacted]"[redacted]"[redacted]
```

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted]——[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted] [redacted] [redacted]

[redacted]

[redacted]——[redacted]

[redacted]

```
  [redacted]
  [redacted]
  [redacted]
  [redacted]
  [redacted]
  [redacted]
  |
  [redacted]
  [redacted]
  [redacted]
  [redacted]
  [redacted]
  [redacted]
  |
  [redacted]
```

[redacted]

```
def [redacted]():
    [redacted]
    [redacted]
    [redacted]
    [redacted]
    [redacted]
    |
    [redacted]
    [redacted]
    [redacted]
```

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

```
IndexError: list index out of range
```

[redacted]panic[redacted]

[redacted]Index[redacted]out of range[redacted]

[redacted]10[redacted]11[redacted]

[redacted]100[redacted]101[redacted]

[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]

[redacted]365[redacted]365[redacted]

```
if [redacted]():
    [redacted]()
elif [redacted]():
    [redacted]()
elif [redacted]():
    [redacted]()
```

[redacted]

- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted]

[redacted]

[redacted]——[redacted]——[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted]

[redacted]——[redacted]

[redacted]

[redacted]

[redacted] [redacted]

x=

CHAPTER 04

資料與變數

程式如何儲存與處理資訊

04

[redacted]

[redacted]

[redacted]...[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

```
[redacted]
[redacted]
```

[redacted]

[redacted]

[redacted]——[redacted]

```
[redacted] = "[redacted]"
[redacted]...
[redacted] = "[redacted]"
[redacted]
```

[redacted]Photoshop[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

```
[redacted] = 3
[redacted] = 50000
[redacted] = 23.5
```

[redacted]

[redacted]3[redacted]3.5[redacted]

- [redacted]Integer[redacted]
- [redacted]Float[redacted]

[redacted]2.5[redacted]25.5[redacted]

[redacted]

[redacted]

```
[redacted] = "[redacted]"
[redacted] = "[redacted]"
[redacted] = "[redacted]1%"
```

[redacted]String[redacted]

[redacted]

[redacted]

[redacted]

```
print(Hello)    # [redacted]Hello[redacted]
print("Hello")  # [redacted]Hello[redacted]
```

[redacted] [redacted]

[redacted]

[redacted]

[redacted]

```
[redacted] = True
[redacted] = False
[redacted]VIP = True
```

[redacted]Boolean[redacted]True[redacted]False[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]

[redacted]

```
[redacted] = ([redacted] >= 1000)
[redacted] = [redacted] and [redacted]
[redacted] = ([redacted] < 10)
```

[redacted]

[redacted]

```
[redacted] = ["[redacted]", "[redacted]", "[redacted]", "[redacted]"]
[redacted] = [120000, 98000, 150000, 87000]
[redacted] = ["[redacted]", "[redacted]", "[redacted]"]
```

[redacted]List[redacted]

[redacted]

```
[redacted] = [
    "[redacted]",
    "[redacted]",
    "[redacted]",
    "[redacted]",
    "[redacted]",
    "[redacted]"
]
```

[redacted]

- [redacted]append[redacted]
- [redacted]remove[redacted]
- [redacted]sort[redacted]
- [redacted]

[redacted]

[redacted]

[redacted]a[redacted]b[redacted]c[redacted]

[redacted]

[redacted]

```
# [redacted]
[redacted] = 50000
[redacted] = 0.1
[redacted] = a * b

# [redacted]
[redacted] = 50000
[redacted] = 0
[redacted] = [redacted] * [redacted]
```

[redacted]

[redacted]

[redacted]

[redacted]2[redacted]

[redacted]

[redacted]

[redacted]

```
# [redacted]
[redacted] = "[redacted]"
```

[redacted]

```
[redacted] = 1000
[redacted] = [redacted] + 500  # [redacted]
[redacted] = [redacted] * 0.9  # [redacted]
```

[redacted]

[redacted]

```
def [redacted]():
    [redacted] = 0.8  # [redacted]
    return [redacted] * [redacted]

# [redacted]
print([redacted])  # [redacted]
```

[redacted]

[redacted]

[redacted]

[redacted]

```
[redacted] = input("[redacted]")  # [redacted]10
[redacted] = 100
[redacted] = [redacted] * [redacted]
```

[redacted]1000[redacted]

[redacted]`input()`[redacted]10[redacted]10[redacted]

```
"10" * 100  # [redacted]
10 * 100    # [redacted]1000
```

[redacted]

```
[redacted] = int(input("[redacted]"))  # [redacted]
```

[redacted]

[redacted]cooking[redacted]

[redacted]

[redacted]

[redacted]

```
[redacted]
[redacted] = "[redacted]"
[redacted] = 3
[redacted] = "[redacted]"

[redacted] = f"{[redacted]} {[redacted]} {[redacted]}"
[redacted]("[redacted]35[redacted]")
```

[redacted]

```
[redacted]
[redacted] = 1000
[redacted] = "[redacted]" + str([redacted]) + "[redacted]"

[redacted]
[redacted] = "25"
[redacted] = int([redacted])

[redacted]
[redacted] = bool([redacted])  # 0[redacted]False[redacted]True
```

[redacted]

```
def [redacted]email(email):
    [redacted] = "@" in email
    [redacted] = "." in email
    return [redacted] and [redacted]

[redacted]email = "wang@example.com"
if [redacted]email([redacted]email):
    [redacted](
else:
    [redacted]("Email[redacted]")
```

[redacted]

[redacted]

[redacted]

[redacted]

```
[redacted]
[redacted] = [
    {
        "[redacted]": 30000,
        "[redacted]": "2024-06-01",
        "[redacted]": "[redacted]",
        "[redacted]": 15000
    },
]

[redacted]
[redacted] = sum([redacted]) / len([redacted])
[redacted] = max([redacted])
[redacted] = [redacted] - [redacted]
```

[redacted]

[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]

[redacted]…[redacted]

[redacted]

[redacted]——[redacted]

[redacted]

[redacted] [redacted]

[redacted]

■Data is the new oil.■ ■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■

CHAPTER 05

自動化你的生活

用程式解決日常問題

05

[redacted]

[redacted]...[redacted]

[redacted]40[redacted]...[redacted]

[redacted]

[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted]

1. [redacted]20[redacted]
2. [redacted]Excel[redacted]
3. [redacted]
4. [redacted]
5. [redacted]
6. [redacted]
7. [redacted]PDF
8. [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]200[redacted]

- IMG_001.jpg
- [redacted]1.png
- [redacted]-2.jpeg
- Copy of DSC1234.JPG

[redacted]

- [redacted]_001.jpg
- [redacted]_002.jpg

- [redacted]_003.jpg
- ...

[redacted]

[redacted]

```
import os

[redacted] = "[redacted]"
[redacted] = os.listdir([redacted])
[redacted] = 1

for [redacted] in [redacted]:
    if [redacted].endswith(('.jpg', '.jpeg', '.png')):
        [redacted] = f"[redacted]{[redacted]:03d}.jpg"
        os.rename([redacted], [redacted])
        [redacted] += 1
        print(f"[redacted]{[redacted]} → {[redacted]}")
```

[redacted]200[redacted]

[redacted]

[redacted]

[redacted]

Excel[redacted]

[redacted]

[redacted]

```
import pandas as pd
import glob

# [redacted]
[redacted] = glob.glob("[redacted]/*.xlsx")

# [redacted]
[redacted] = pd.DataFrame()

```

```
# [redacted]
for [redacted] in [redacted]:
    [redacted] = [redacted].split("/")[-1].replace(".xlsx", "")

    # [redacted]Excel
    [redacted] = pd.read_excel([redacted])

    # [redacted]
    [redacted]["[redacted]"] = [redacted]

    # [redacted]
    [redacted] = pd.concat([[redacted], [redacted]])

# [redacted]
[redacted] = [redacted]["[redacted]"].sum()
[redacted] = [redacted]["[redacted]"].mean()
[redacted] = [redacted].groupby("[redacted]")["[redacted]"].sum().idxmax()

# [redacted]
[redacted].to_excel("[redacted].xlsx", index=False)
print(f"[redacted]{[redacted]:,.0f}")
print(f"[redacted]{[redacted]:.2f}")
print(f"[redacted]{[redacted]}")
```

[redacted]...[redacted]10[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

1. [redacted]
2. [redacted]
3. [redacted]
4. [redacted]
5. [redacted]
6. [redacted]

7. [redacted]50[redacted]

[redacted]

```
import smtplib
from email.mime.text import MIMEText
from datetime import datetime

def [redacted]([redacted], email, [redacted], [redacted]):
    [redacted] = f"[redacted] - {[redacted]}"

    [redacted] = f""
    [redacted]{[redacted]}"

    [redacted] ${[redacted]:,}

    [redacted]{[redacted]

    [redacted]

    [redacted]
    [redacted]
    """

    # [redacted]
    [redacted](email, [redacted], [redacted])
    print(f"[redacted] {[redacted]}")

    [redacted]
    [redacted] = pd.read_excel("[redacted].xlsx")

    for _, [redacted] in [redacted].iterrows():
        [redacted](
            [redacted]["[redacted]"],
            [redacted]["Email"],
            [redacted]["[redacted]"],
            [redacted]["[redacted]"]
        )
```

50[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

1. [redacted]10[redacted]
2. [redacted]
3. [redacted]
4. [redacted]Excel
5. [redacted]

[redacted]

[redacted]

```
import requests
from bs4 import BeautifulSoup
import schedule

def [redacted]():
    [redacted] = [
        "https://competitor1.com/product",
        "https://competitor2.com/product",
        # ...
    ]

    [redacted] = {}

    for [redacted] in [redacted]:
        [redacted] = requests.get([redacted])
        [redacted] = BeautifulSoup([redacted].text, 'html.parser')

        # [redacted]
        [redacted] = [redacted].find('span', class_='price')
        [redacted] = float([redacted].text.replace('$', ''))

        [redacted] = [redacted].split('/')[2]
        [redacted][[redacted]] = [redacted]

    # [redacted]Excel
    [redacted]([redacted])

    # [redacted]
    for [redacted], [redacted] in [redacted].items():
        if [redacted]10%:
            [redacted]f"{[redacted]}[redacted]"

    [redacted]9[redacted]
schedule.every().day.at("09:00").do([redacted])
```

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

```
import os
import shutil
from datetime import datetime

def [redacted]():
    [redacted] = os.path.expanduser("~/Desktop")

    # [redacted]
    [redacted] = {
        "[redacted]": [".jpg", ".png", ".gif", ".psd"],
        "[redacted]": [".doc", ".docx", ".pdf", ".txt"],
        "[redacted]": [".xls", ".xlsx", ".csv"],
        "[redacted]": [".mp4", ".mov", ".avi"],
        "[redacted]": []
    }

    # [redacted]
    for [redacted] in [redacted].keys():
        [redacted] = os.path.join([redacted], [redacted])
        if not os.path.exists([redacted]):
            os.makedirs([redacted])

    # [redacted]
    for [redacted] in os.listdir([redacted]):
        if os.path.isfile(os.path.join([redacted], [redacted])):
            [redacted] = "[redacted]"

            for [redacted], [redacted] in [redacted].items():
                if any([redacted].endswith(ext) for ext in [redacted]):
                    [redacted] = [redacted]
                    break

            # [redacted]
            [redacted] = os.path.join([redacted], [redacted])
            [redacted] = os.path.join([redacted], [redacted], [redacted])
            shutil.move([redacted], [redacted])
            print(f"[redacted] {[redacted]} → {[redacted]}/")
```

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

```
import shutil
import schedule
from datetime import datetime

def [redacted]():
    [redacted] = [
        "[redacted]",
        "[redacted]",
        "[redacted]",
    ]

    [redacted] = datetime.now().strftime("%Y%m%d")
    [redacted] = f"[redacted]_{[redacted]}"

    for [redacted] in [redacted]:
        [redacted] = f"C:/Users/[redacted]/{[redacted]}"
        [redacted] = f"D:/[redacted]/{[redacted]}/{[redacted]}"

        shutil.copytree([redacted], [redacted])
        print(f"[redacted]{[redacted]}")

    # [redacted] 30[redacted]
    [redacted]()

# [redacted]
schedule.every().day.at("17:30").do([redacted])

# [redacted]
schedule.every().friday.at("18:00").do([redacted])
```

[redacted]

[redacted]

[redacted]

[redacted]

- [redacted] → RSS[redacted]
- [redacted] → [redacted]
- [redacted] → [redacted]
- [redacted] → [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted] [redacted] [redacted] [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

■■■■■■

■*The first rule of any technology used in a business is that automation applied to an efficient operation will magnify the efficiency.*■ *- Bill Gates*

■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■

CHAPTER 06

創造你的第一個專案

06

動手實作，從想法到成品

[redacted]

[redacted]...[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]——[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]App[redacted]

[redacted]

[redacted]

[redacted]

[redacted]——[redacted]

[redacted]

```
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]

[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
```

[redacted]

[redacted]

[redacted]

```
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
```

```
[redacted]
```

[redacted]

[redacted]

[redacted]

```
import [redacted]
import [redacted]
from typing import Dict, List

class [redacted]:
    def __init__(self):
        self.[redacted] = []
        self.[redacted] = [
            "[redacted]", "[redacted]", "[redacted]", "[redacted]",
            "[redacted]", "[redacted]", "[redacted]", "[redacted]"
        ]
        self.[redacted] = 30000
        self.[redacted]()

    def [redacted](self):
        """[redacted]"""
        try:
            with open("[redacted].json", "r", encoding="utf-8") as f:
                [redacted] = json.load(f)
                self.[redacted] = [redacted].get("[redacted]", [])
                self.[redacted] = [redacted].get("[redacted]", 30000)
        except FileNotFoundError:
            print("[redacted]")

    def [redacted](self):
        """[redacted]"""
        [redacted] = {
            "[redacted]": self.[redacted],
            "[redacted]": self.[redacted]
```

```
    }
    with open("[redacted].json", "w", encoding="utf-8") as f:
        json.dump([redacted], f, ensure_ascii=False, indent=2)
```

[redacted]

[redacted]

[redacted]——[redacted]

[redacted]

```
def [redacted](self, [redacted]: float, [redacted]: str, [redacted]: str = ""):
    """[redacted]"""
    if [redacted] not in self.[redacted]:
        print(f"[redacted] '{[redacted]}' [redacted]")
        [redacted] = "[redacted]"

    [redacted] = {
        "[redacted]": datetime.datetime.now().strftime("%Y-%m-%d"),
        "[redacted]": datetime.datetime.now().strftime("%H:%M"),
        "[redacted]": "[redacted]",
        "[redacted]": [redacted],
        "[redacted]": [redacted],
        "[redacted]": [redacted]
    },

    self.[redacted].append([redacted])
    self.[redacted]()

    # [redacted]
    self.[redacted]()

    print(f"✓ [redacted]{[redacted]} -${[redacted]}")

def [redacted](self, [redacted]: float, [redacted]: str = "[redacted]"):
    """[redacted]"""
    [redacted] = {
        "[redacted]": datetime.datetime.now().strftime("%Y-%m-%d"),
        "[redacted]": datetime.datetime.now().strftime("%H:%M"),
        "[redacted]": "[redacted]",
        "[redacted]": [redacted],
        "[redacted]": [redacted],
        "[redacted]": [redacted]
    },

    self.[redacted].append([redacted])
    self.[redacted]()
```

```
[redacted]
    print(f"✓ [redacted]{[redacted]} +${[redacted]}")
```

[redacted]

[redacted]

[redacted]

```
def [redacted](self):
    """[redacted]"""
    [redacted] = datetime.datetime.now().strftime("%Y-%m")
[redacted]
    [redacted] = []
    [redacted] for [redacted] in self.[redacted]
        if [redacted]["[redacted]"].startswith([redacted])
    [redacted]
[redacted]
    [redacted] = sum(
        [redacted]["[redacted]"] for [redacted] in [redacted]
        if [redacted]["[redacted]"] == "[redacted]"
    )
[redacted]
    [redacted] = sum(
        [redacted]["[redacted]"] for [redacted] in [redacted]
        if [redacted]["[redacted]"] == "[redacted]"
    )
[redacted]
    [redacted] = [redacted] - [redacted]
[redacted]
    print(f"\n[redacted] {[redacted]} [redacted]")
    print("=" * 40)
    print(f"[redacted]${[redacted]:,.0f}")
    print(f"[redacted]${[redacted]:,.0f}")
    print(f"[redacted]  ${[redacted]:,.0f}")
[redacted]
    # [redacted]
    [redacted] = {}
    for [redacted] in [redacted]:
        if [redacted]["[redacted]"] == "[redacted]":
            [redacted] = [redacted]["[redacted]"]
            [redacted][[redacted]] = [redacted].get([redacted], 0) + [redacted]["[redacted]"]
[redacted]
    if [redacted]:
        print("\n[redacted]")
        for [redacted], [redacted] in sorted([redacted].items(), key=lambda x: x[1], reverse=True):
            [redacted] = ([redacted] / [redacted]) * 100
            [redacted] = "[redacted]" * int([redacted] / 5)
```

```
            print(f"  {[illegible]:<4} ${[illegible]:>8,.0f} {[illegible]} {[illegible]:.1f}%")
```

[illegible]40%[illegible]

[illegible]

[illegible]

```
def [illegible](self, [illegible]: float):
    """[illegible]"""
    self.[illegible] = [illegible]
    self.[illegible]()
    print(f"✓ [illegible]${[illegible]:,.0f}")

def [illegible](self):
    """[illegible]"""
    [illegible] = datetime.datetime.now().strftime("%Y-%m")

    [illegible] = sum(
        [illegible]["[illegible]"] for [illegible] in self.[illegible]
        if [illegible]["[illegible]"].startswith([illegible]) and [illegible]["[illegible]"] == "[illegible]"
    )

    [illegible] = ([illegible] / self.[illegible]) * 100
    [illegible] = self.[illegible] - [illegible]

    if [illegible] > 100:
        print(f"[illegible] [illegible] ${-[illegible]:,.0f}[illegible]")
    elif [illegible] > 80:
        print(f"[illegible] [illegible] {[illegible]:.0f}%[illegible] ${[illegible]:,.0f}")
    elif [illegible] > 50:
        print(f"[illegible] [illegible] {[illegible]:.0f}%[illegible] ${[illegible]:,.0f}")
```

[illegible]3000[illegible]500[illegible]

[illegible]

[illegible]

```
def [illegible](self):
    [illegible]"""
```

```
    while True:
        print("\n" + "=" * 40)
        print("[redacted]")
        print("=" * 40)
        print("1. [redacted]")
        print("2. [redacted]")
        print("3. [redacted]")
        print("4. [redacted]")
        print("5. [redacted]")
        print("6. [redacted]")
        print("0. [redacted]")

        [redacted] = input("\n[redacted]")

        if [redacted] == "1":
            self.[redacted]()
        elif [redacted] == "2":
            self.[redacted]()
        elif [redacted] == "3":
            self.[redacted]()
        elif [redacted] == "4":
            self.[redacted]()
        elif [redacted] == "5":
            self.[redacted]()
        elif [redacted] == "6":
            self.[redacted]()
        elif [redacted] == "0":
            print("[redacted]")
            break
        else:
            print("[redacted]")

def [redacted](self):
    """[redacted]"""
    print("\n[redacted]")
    print("-" * 20)

    # [redacted]
    print("[redacted]")
    for i, [redacted] in enumerate(self.[redacted], 1):
        print(f"  {i}. {[redacted]}")

    [redacted] = input("[redacted]")
    try:
        [redacted] = self.[redacted][int([redacted]) - 1]
    except:
        [redacted] = "[redacted]"

    [redacted] = float(input("[redacted]$"))
    [redacted] = input("[redacted]")

    self.[redacted]([redacted], [redacted], [redacted]
```

[redacted]

[redacted]

[redacted]

[redacted]

[redacted] [redacted] [redacted]

[redacted]——[redacted]

```
def [redacted](self):
    """[redacted]"""
    if not self.[redacted]:
        print("[redacted]")
        return

    [redacted] = self.[redacted].pop()
    self.[redacted]()

    print(f"✓ [redacted]{[redacted]['[redacted]']} {[redacted]['[redacted]']} ${[redacted]['[redacted]']}")

def [redacted](self, [redacted]: str):
    """[redacted]"""
    [redacted] = []
        [redacted] for [redacted] in self.[redacted]
        if [redacted] in [redacted].get('[redacted]', "") or [redacted] in [redacted]['[redacted]']
    ]

    if [redacted]:
        for [redacted] in [redacted]:
            print(f"{[redacted]['[redacted]']} {[redacted]['[redacted]']} ${[redacted]['[redacted]']} {[redacted].get('[redacted]', '')}")
    else:
        print(f"[redacted] '{[redacted]}' [redacted]")
```

[redacted]

[redacted]

[redacted]

[redacted]——[redacted]

```
# [redacted]
# requirements.txt
pandas==1.5.0
matplotlib==3.6.0

# [redacted]
[redacted]
[redacted]
1. [redacted]Python 3.8[redacted]
2. [redacted]pip install -r requirements.txt
3. [redacted]python [redacted].py

# [redacted]
pyinstaller --onefile [redacted].py
[redacted]
```

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]code[redacted]

[redacted]

[redacted]

[redacted]——[redacted]

[redacted]

[redacted]

[redacted]

- [redacted]
- [redacted]

- ■■■■■App
- ■■■■■■■■■■

■■■■■■■■■■■■■■■■■■■■■■

■■■■■■■■■■■■■■■■■■■■■■■■■■■■

■■■■■■■■■■■——■■■■■■■■■

■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■

■■■■■■■■■

■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■

■■■■■■■■■■■■■■■■■■■■■■■■■■■——■■■■■■

■■■■■■■■■■■■■■■■■■

■*The best way to predict the future is to invent it.*■ *- Alan Kay*

■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■

//

CHAPTER 07

程式設計的職業道路

了解產業與職涯發展

07

7

[redacted]

[redacted]35[redacted]

[redacted]

[redacted]

[redacted]——[redacted]

[redacted]28[redacted]

[redacted]

[redacted]

[redacted]...

[redacted]

```
[redacted] ← [redacted] →
│
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
│
[redacted]
[redacted]
iOS / Android
[redacted] App
│
[redacted] DevOps [redacted]
[redacted]
[redacted]
[redacted]
│
[redacted]
[redacted]
[redacted]
[redacted] ETL [redacted]
│
[redacted] AI/ML [redacted]
[redacted]
[redacted]
[redacted]
```

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

- [redacted] standup meeting[redacted]
- [redacted]
- [redacted]

- [redacted]
- [redacted]
- [redacted] API [redacted]

[redacted]

```
├── [redacted]
│   ├── HTML [redacted]
│   ├── CSS [redacted]
│   └── JavaScript [redacted]
│
├── [redacted]
│   └── React / Vue / Angular [redacted]
│
└── [redacted]
    ├── Git [redacted]
    ├── Webpack [redacted]
    └── npm [redacted]
```

[redacted]

- [redacted]1px [redacted]
- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

- [redacted]
- [redacted] API [redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]

[redacted]

```
[redacted]
Python + Django/Flask
JavaScript + Node.js
Java + Spring
Go
│
[redacted]
MySQL/PostgreSQL
NoSQL: MongoDB, Redis
│
[redacted]
RESTful API
[redacted]
[redacted]
[redacted]
```

[redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

- [redacted]ETL[redacted]
- [redacted]
- [redacted]
- [redacted]
- [redacted]

[redacted]

- [redacted]
- [redacted]

- [redacted]

[redacted]Excel[redacted]

[redacted]

[redacted]

[redacted]

[redacted]0-3[redacted]

```
[redacted]
[redacted]
[redacted]
[redacted]Git [redacted]
│
[redacted]2-3[redacted]
[redacted]
```

[redacted]4-6[redacted]

```
[redacted]
[redacted]JavaScript
[redacted]React/Vue[redacted]
[redacted]3[redacted]
│
[redacted]
[redacted]
[redacted]
[redacted]RESTful API[redacted]
```

[redacted]7-9[redacted]

```
[redacted]
[redacted]1-2[redacted]
[redacted]GitHub
[redacted]
[redacted]
```

[redacted]10-12[redacted]

```
[redacted]
[redacted] LeetCode[redacted]
[redacted]
[redacted]
[redacted]
```

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

1. [redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]

2. [redacted]/[redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]

3. [redacted]

- [redacted]
- [redacted]
- [redacted]

4. [redacted]

- [redacted]
- [redacted]
- [redacted]

[redacted]...[redacted]

[redacted]

[redacted]

[redacted]

```
[redacted]2024[redacted]
│
junior[redacted]0-2[redacted]
[redacted]40-60[redacted]/[redacted]
[redacted]45-65[redacted]/[redacted]
[redacted]45-65[redacted]7[redacted]
│
mid-level[redacted]2-5[redacted]
[redacted]60-90[redacted]/[redacted]
[redacted]65-100[redacted]/[redacted]
[redacted]70-100[redacted]7[redacted]
│
senior[redacted]5[redacted]
[redacted]90-150[redacted]/[redacted]
[redacted]100-180[redacted]/[redacted]
[redacted]100-180[redacted]7[redacted]
│
[redacted]/[redacted]
[redacted]
```

[redacted]

[redacted]

[redacted]

[redacted]

```
[redacted]
[redacted] code review [redacted]
[redacted]15[redacted]
[redacted]
[redacted] debug
[redacted]
[redacted]
[redacted]
[redacted] Code review [redacted]
[redacted]/[redacted]
[redacted]

[redacted]
[redacted] 30% [redacted]
[redacted] 30% [redacted]/debug
[redacted] 20% [redacted]
[redacted] 20% [redacted]/[redacted]
```

[redacted]

[redacted]

[redacted]

[redacted]

```
[redacted] → [redacted]
[redacted] → 6[redacted]

[redacted]
jQuery[redacted]→ [redacted]
[redacted] 5 [redacted] → [redacted]
[redacted] → [redacted]
```

[redacted]

[redacted]

[redacted]

[redacted]

[redacted] + [redacted] = [redacted]

- [redacted] + [redacted] = [redacted]
- [redacted] + [redacted] = [redacted]
- [redacted] + [redacted] = FinTech [redacted]
- [redacted] + [redacted] = [redacted]

[redacted] [redacted]
[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

■*The expert in anything was once a beginner.*■

■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■

CHAPTER 08

學習資源與社群

找到持續學習的方法

08

8

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]——YouTube[redacted]GitHub[redacted]Medium[redacted]

[redacted]——[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

YouTube[redacted]

[redacted]

[redacted]

freeCodeCamp

- [redacted]4-8[redacted]
- [redacted]
- [redacted]

Traversy Media

- [redacted]
- [redacted]
- [redacted]Brad[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]

[redacted]YouTube[redacted]——[redacted]

[redacted]

GitHub[redacted]

[redacted]GitHub[redacted]

[redacted]

[redacted]GitHub[redacted]

1. [redacted]

```
[redacted]
[redacted] "awesome" [redacted]
[redacted]
[redacted] Trending [redacted]
```

2. [redacted]README[redacted] [redacted]README[redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]

3. [redacted] [redacted]

- developer-roadmap[redacted]
- free-programming-books[redacted]
- project-based-learning[redacted]

[redacted]GitHub Trending[redacted]

[redacted]

Codecademy[redacted]

```
[redacted]
[redacted]
[redacted]
[redacted]
|
[redacted]
[redacted]
[redacted]
|
[redacted]
[redacted]
[redacted]
```

HackerRank / LeetCode[redacted]

```
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted] Easy [redacted]
[redacted]
[redacted]
```

[redacted]

[redacted]

[redacted]

Udemy[redacted]

```
[redacted]
[redacted]
[redacted]
[redacted]
[redacted] $300-500 [redacted]
[redacted]
[redacted]
[redacted] 4.5 [redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted] React [redacted]
[redacted]
[redacted]
[redacted] YouTube [redacted]
```

Coursera[redacted]

```
[redacted]
[redacted]
[redacted]
```

```
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]Princeton[redacted]
[redacted]
```

[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]

Hahow

- [redacted]
- [redacted]
- [redacted]

Alpha Camp

- [redacted]
- [redacted]mentor[redacted]
- [redacted]

[redacted]

[redacted]bug[redacted]

Facebook[redacted]

[redacted]

[redacted]JavaScript.tw[redacted]

- [redacted]3[redacted]+
- [redacted]
- [redacted]

[redacted]Python Taiwan[redacted]

- [redacted]2[redacted]+
- [redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]

[redacted]

1. [redacted]
2. [redacted]
3. [redacted]
4. [redacted]

Discord / Slack

[redacted]

[redacted]Discord[redacted]

[redacted]

[redacted]

Meetup

[redacted]

- ReactJS Taiwan
- Python Taipei
- GDG Taiwan

[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

1. [redacted]
2. [redacted]
3. [redacted]
4. [redacted]other beginners

[redacted]

```
[redacted]
[redacted]
[redacted]bug
[redacted]
[redacted]
|
[redacted]
[redacted]
```

[redacted]

Medium

- [redacted]
- [redacted]
- [redacted]

dev.to

- [redacted]
- [redacted]
- [redacted]

iT[redacted]

- [redacted]
- [redacted]

- [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

JavaScript Jabber [redacted]

CodeNewbie [redacted]

[redacted] [redacted]

[redacted]

Twitter/X [redacted]

[redacted]timeline[redacted]

```
[redacted]
  Dan Abramov (React[redacted])
  Addy Osmani (Google Chrome)
  Sara Soueidan (CSS[redacted])
[redacted]
  [redacted]
  [redacted]Kon
  [redacted]
```

[redacted]

Marie Kondo[redacted]

[redacted]

[redacted]

```
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
```

[redacted]

```
[redacted]
[redacted]
|
[redacted]
[redacted]
|
[redacted]
[redacted]
```

[redacted]300[redacted]30[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]——[redacted]

[redacted]

[redacted]*The capacity to learn is a gift; The ability to learn is a skill; The willingness to learn is a choice.*[redacted]

[redacted] [redacted] [redacted]

[redacted]

CHAPTER 09

學習程式的正確心態

09

克服挫折，建立成長思維

9

bug

Google Stack Overflow

[redacted]bug[redacted]

[redacted]——[redacted]

[redacted]

---

bug[redacted]

[redacted]

[redacted]

[redacted]

[redacted]——[redacted]`getElementById` [redacted] `getElementByID`[redacted]ID[redacted]

[redacted]

[redacted]

[redacted]

[redacted]——JavaScript[redacted]

[redacted]bug[redacted]bug[redacted]

```
bug[redacted]
│
[redacted] → [redacted]
[redacted] → [redacted]
[redacted] → [redacted]
[redacted] → [redacted]
│
[redacted]bug[redacted]
[redacted]
```

[redacted]bug[redacted]bug[redacted]

[redacted]

[redacted]

[redacted]

[redacted]Impostor Syndrome[redacted]——[redacted]

[redacted]

[redacted]

- 70%[redacted]
- 58%[redacted]
- [redacted]

[redacted]

[redacted]

[redacted]

1. [redacted]

[redacted]

[redacted]

```
[redacted]
[redacted]
[redacted]
[redacted]

[redacted]
[redacted]Python[redacted]
[redacted]5[redacted]
[redacted]8[redacted]
```

2. [redacted]

```
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
```

```
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]budget[redacted]
[redacted]
```

[redacted]

3. [redacted]

[redacted]

[redacted]50[redacted]20[redacted]

[redacted]

[redacted]

```
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
```

[redacted]

[redacted]

[redacted]

[redacted]

1. [redacted]

[redacted]CRUD[redacted]

- [redacted]
- [redacted]
- [redacted]

2. [redacted]

[redacted]

3. [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

Done is better than perfect

Facebook[redacted]Done is better than perfect[redacted]

```
[redacted]
|
[redacted] → [redacted]
[redacted] → [redacted]
[redacted] → [redacted]
[redacted] → [redacted]
```

[redacted]

1. [redacted]
2. [redacted]
3. [redacted]
4. [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

- [redacted]React
- [redacted]Vue
- [redacted]Svelte

[redacted]

[redacted]

```
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
[redacted]
```

FOMO vs JOMO

FOMO■Fear of Missing Out■■■■■■ JOMO■Joy of Missing Out■■■■■■

■■■■■■

[redacted]

[redacted]bug[redacted]

[redacted]

[redacted]

[redacted]50[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

- [redacted]bug → [redacted]smile
- [redacted] → [redacted]
- [redacted] → [redacted]

[redacted]

```
[redacted]
[redacted]
[redacted]bug
[redacted]
[redacted]
```

[redacted]

[redacted]

[redacted]

[redacted]

- [redacted]bug
- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted]admire[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]*The master has failed more times than the beginner has even tried.*[redacted]

[redacted]

[redacted] [redacted]

CHAPTER 10

從零到一的行動計畫

制定你的學習路線圖

10

[redacted]10[redacted]

[redacted]...[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]50[redacted]Python[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]——[redacted]

```
[redacted]
[redacted]/[redacted] → Python
[redacted]
│
[redacted]/[redacted] → JavaScript
[redacted]
│
[redacted]/[redacted] → Python
[redacted]
│
[redacted] → JavaScript [redacted] Python
[redacted]
│
[redacted] → Python
[redacted]
```

[redacted]

1. [redacted]Python[redacted]python.org[redacted]Node.js[redacted]nodejs.org[redacted]
2. [redacted]VS Code[redacted]code.visualstudio.com[redacted]
3. [redacted]`print("[redacted]")`

[redacted]

[redacted]

[redacted]

[redacted]8[redacted]

■■30■■■■■

```
■■■■■■■■■■■
■■■■■■ = "■■■■■"
    ■■ = ■■■■■
    print(f"■■{■■}■■■■{■■}■■")

■■■■■■■■
■■■■■■■■ >= 18:
        print("■■■■")
    ■■■
        print("■■■■")

■■■■■■
■■■for ■■ in range(10):
        print(f"■■{■■+1}■■■■")

■■■■■■
■■■def ■■(■■):
        return f"■■■■{■■}■■"

■■■■■■■■
■■■■■■■■■■■■
■■■■■■■■■■■■
```

■■■■■■■■■■■■■■■■■■10■■■■

■■■■■■■■■

■■■■■■■■■■

■1■■■■■■■

- ■■■■■■■■■■■■■■■
- ■■■■■■■■■■■■■■
- ■■■■■■■■■■■

■2■■■■■■■

- ■■■■■■■■■■■■■■■
- ■■■■■■■■■■CSV■■

- [redacted]

[redacted]3[redacted]

- [redacted]
- [redacted]App
- [redacted]

[redacted]4[redacted]

- [redacted]
- [redacted]
- [redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted]

[redacted]2[redacted]

```
[redacted]
[redacted]
[redacted]
- HTML/CSS[redacted]
- JavaScript DOM[redacted]
- [redacted]
[redacted]
[redacted]
- pandas[redacted]
- [redacted]
```

```
Excel
```

3

3-5

-
-
- GitHub
-

A

Python

-
-
-
-

■■B■■■■■■

■■■■■■■■■T■■■■

■■■■■■■■■■■■■■■■■■

■■■■■

- ■■■■■■
- ■■■■■
- ■■■■■■
- ■■■■■■

■■C■■■■■■

■■■■■■■■■■■■

■■■■■■■■■■■■

■■■■■

- ■■■■4■■■■
- ■■■■■■■■
- ■LeetCode■■
- ■■■■

■■■■■■■■■■

■■■■

■■■■■■■■■■■■■■■

■■1■■■■■

```
[illegible]100[illegible]
[illegible]
```

[illegible]2[illegible]

```
[illegible]
[illegible]
```

[illegible]3[illegible]

```
[illegible]
[illegible]
```

[illegible]4[illegible]

```
[illegible]
[illegible]
```

[illegible]5[illegible]

```
[illegible]5[illegible]
[illegible]
```

[illegible]

[illegible]

[illegible]

```
def [illegible]():
    # [illegible]5[illegible]
    [illegible]()

    # [illegible]15[illegible]
    [illegible]()

    # [illegible]20[illegible]
```

```
    [redacted]()
[redacted]
    # [redacted]30[redacted]
    [redacted]()
[redacted]
    # [redacted]10[redacted]
    [redacted]()
```

[redacted]

[redacted]2-4[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted] [redacted]5:30-6:30[redacted]

[redacted] [redacted]11[redacted]2[redacted]

[redacted] [redacted]12:30-13:30[redacted]

[redacted] [redacted]40[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

- [redacted]10,000[redacted]
- [redacted]10-20[redacted]
- [redacted]bug
- [redacted]
- [redacted]
- [redacted]
- [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted] [redacted] [redacted]Python download[redacted]Node.js download[redacted] [redacted] [redacted] [redacted]

```
print("Hello, World!")
print("[redacted]")
```

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted] [redacted] [redacted] [redacted]

[redacted]——

[redacted]

[redacted]

[redacted]

[redacted]

[redacted] [redacted] [redacted]"Hello World"[redacted]

[redacted]

[redacted] [redacted] [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]55[redacted]

[redacted]

[redacted] [redacted]

[redacted]

[redacted]

[redacted] [redacted] [redacted]

[redacted] [redacted] [redacted]

[redacted]

[redacted]

[redacted]

[redacted]The future belongs to those who learn more skills and combine them in creative ways.[redacted] — Robert Greene

[redacted]

[redacted]

[redacted]

[redacted] [redacted] [redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

- [redacted]
- [redacted]
- [redacted]

- [redacted]

[redacted]

[redacted]

[redacted] Instagram [redacted] ![Instagram QR Code]([redacted]/qr_instagram.png) **@datasunnie** --- ### [redacted] LINE [redacted] [redacted] ![LINE QR Code]([redacted]/qr_line.png) **@055zjitb**

[redacted]

1. [redacted] — [redacted] QR Code
2. [redacted] — [redacted]10[redacted]
3. [redacted] — [redacted]

[redacted] **[redacted] | Data Sunnie** *[redacted]*

[redacted]

[redacted]

[redacted]

[redacted]

— Sunnie